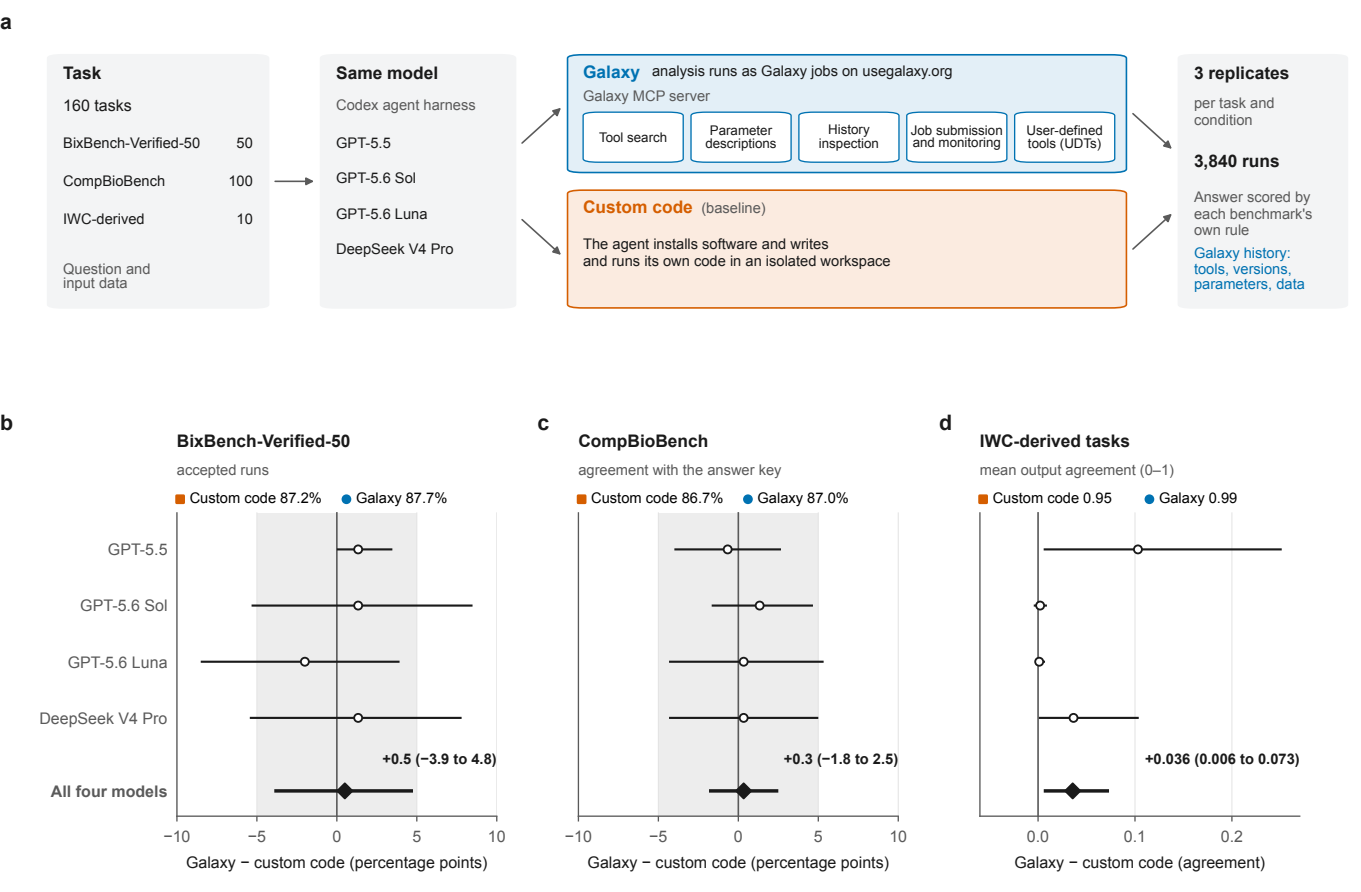

a
Task
160 tasks
BixBench-Verified-50 50
CompBioBench 100
IWC-derived 10
Question and input data
Same model
Codex agent harness
GPT-5.5
GPT-5.6 Sol
GPT-5.6 Luna
DeepSeek V4 Pro
Galaxy analysis runs as Galaxy jobs on usegalaxy.org
Galaxy MCP server
Tool search
Parameter descriptions
History inspection
Job submission and monitoring
User-defined tools (UDTs)
Custom code (baseline)
The agent installs software and writes and runs its own code in an isolated workspace
3 replicates
per task and condition
3,840 runs
Answer scored by each benchmark's own rule
Galaxy history: tools, versions, parameters, data
b
BixBench-Verified-50
accepted runs
Custom code 87.2%
Galaxy 87.7%
GPT-5.5
GPT-5.6 Sol
GPT-5.6 Luna
DeepSeek V4 Pro
+0.5 (−3.9 to 4.8)
All four models
−10 −5 0 5 10
Galaxy − custom code (percentage points)
c
CompBioBench
agreement with the answer key
Custom code 86.7%
Galaxy 87.0%
+0.3 (−1.8 to 2.5)
−10 −5 0 5 10
Galaxy − custom code (percentage points)
d
IWC-derived tasks
mean output agreement (0–1)
Custom code 0.95
Galaxy 0.99
+0.036 (0.006 to 0.073)
0.0 0.1 0.2
Galaxy − custom code (agreement)